ACTIVE SPEAKER – ESP32-WROVER / PCM5122 / TPA3126D2

12–16 V DC -> protected PVDD -> stereo BTL amplifier.
PVDD -> 4.30 V FPWM buck -> ESP32 3.3 V LDO + DAC 3.3 V low-noise LDO.
PCM5122 LEFT = tweeter / J17; RIGHT = woofer / J16. Both speaker returns are active.
C21: onboard 47u / 50V bipolar Nichicon UES; JP1 shorts it for stereo use.
No passive crossover. DSP crossover/EQ/limiting remains a firmware and acoustic task.

POWER

PVDD
4V_PRE
3V3-ESP
VIN_SENSE

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

File: Power.kicad_sch

ESP32

3V3-ESP
VIN_SENSE
BCKD
LRCKD
DATAD
SCLD
SDA
DAC_XSMT
AMP_ENABLE
AMP_MUTE_CMD
AMP_FAULT_N

File: ESP32.kicad_sch

DAC

4V_PRE
BCK
LRCK
DATA
SCL
SDA
DAC_XSMT
WOOFER_AUDIO
TWEETER_AUDIO

File: DAC.kicad_sch

AMP

PVDD
3V3-ESP
WOOFER_AUDIO
TWEETER_AUDIO
AMP_ENABLE
AMP_MUTE_CMD
AMP_FAULT_N

File: Amp.kicad_sch

STARTUP: rails settle -> ESP32 boots (SDZ low, MUTE high, XSMT low).
Configure PCM5122 at 0x4C with both channels muted; configure DSP and valid 64fs I2S.
Release DAC XSMT with valid muted I2S; enable SDZ, wait >=50 ms, release MUTE; ramp volume.
SHUTDOWN: ramp down, assert DAC XSMT low >=150/fs +0.2 ms; assert MUTE then SDZ low.
FAULT: mute and shut down, log fault; deliberate retry only after fault cause is removed.
Review report and bench validation are required before connecting the real drivers.

HS1 Heatsink

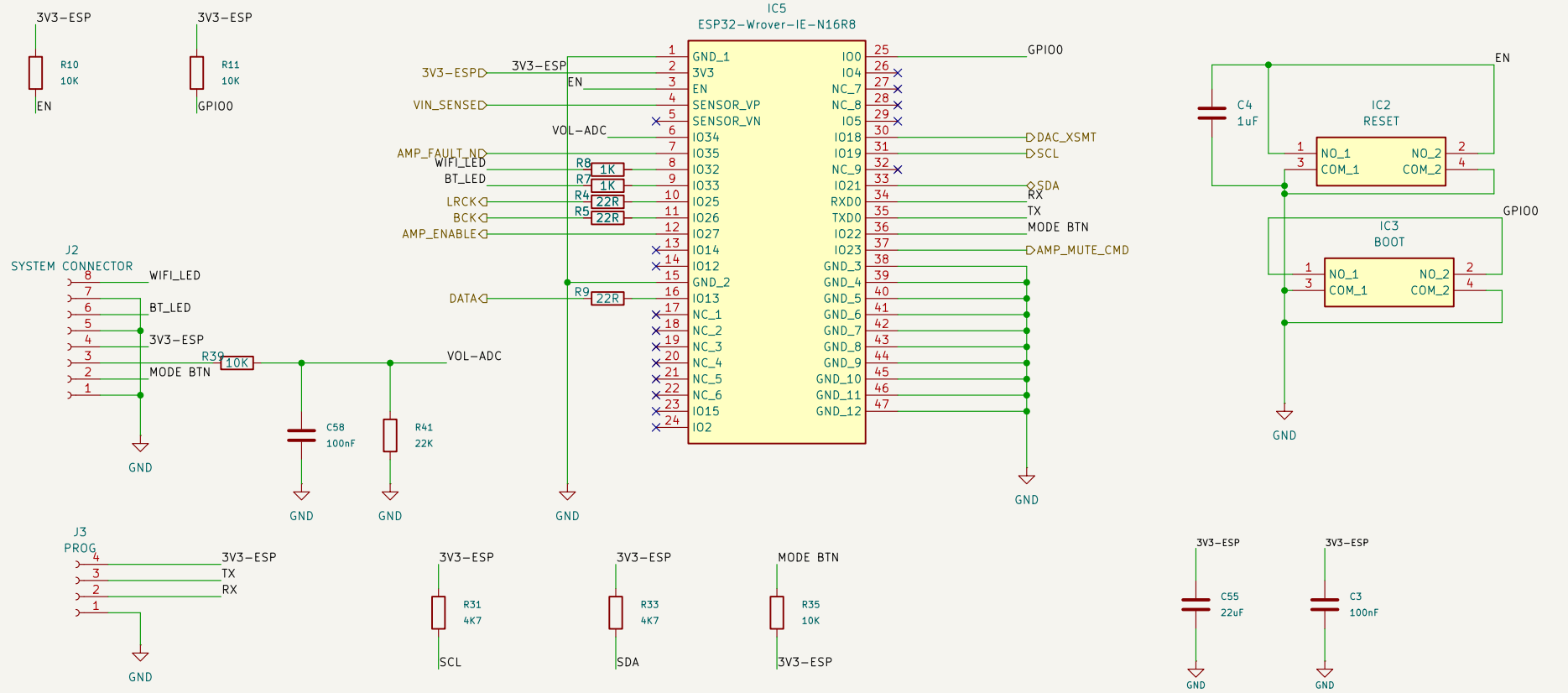
Schematic completion only – PCB unchanged

Sheet: /
File: TPA3126.kicad_sch

Title: Bluetooth / Snapcast active speaker

Size: A3	Date: 2026-10-23	**Rev: Pre-order B**
KiCad E.D.A. 10.0.5		Id: 1/5

ESP32-WROVER-IE-N16R8: EXTERNAL ANTENNA



Preserved: LED Wi-Fi=32, BT=33; LRCK=25, BCK=26, DIN=13; SCL=19, SDA=21.
Added: SDZ enable=27; MUTE command=23 (HIGH=mute); FAULTZ=35 (input only); DAC_XSMT=18.
GPIO36 ADC1 monitors PVDD/11. Volume pot: GPIO34 / ADC1_CH6 (module pin 6). GPIO0 BOOT pull-up; EN reset has 10k/1u RC.
GPIO6-11 / GPIO16-17 reserved for internal flash/PSRAM: no external use.
J2: 1=GND, 2=MODEBTN, 3=pot wiper, 4=3V3, 5=GND.
J2: 6=BT LED drive, 7=GND, 8=Wi-Fi LED drive (both drives via 1k).
MODEBTN: GPIO22 / module pin 36, R35=10k pull-up; switch pin 2 to pin 5 GND.
Debounce in firmware. J3 is 3.3 V UART; module pads 39-47 are GND.
IE module requires an external antenna; keep RF feed away from switch nodes.

VOLUME POT: J2 pins 1 or 5=GND, 3=wiper, 4=3V3. Use 10k linear.
R39/R41 scale full travel to ~2.27V; C58 filters ADC noise.
GPIO34 = ADC1_CH6. Firmware: highest ADC attenuation, average readings;
calibrate endpoints / taper and map low or disconnected wiper to mute.

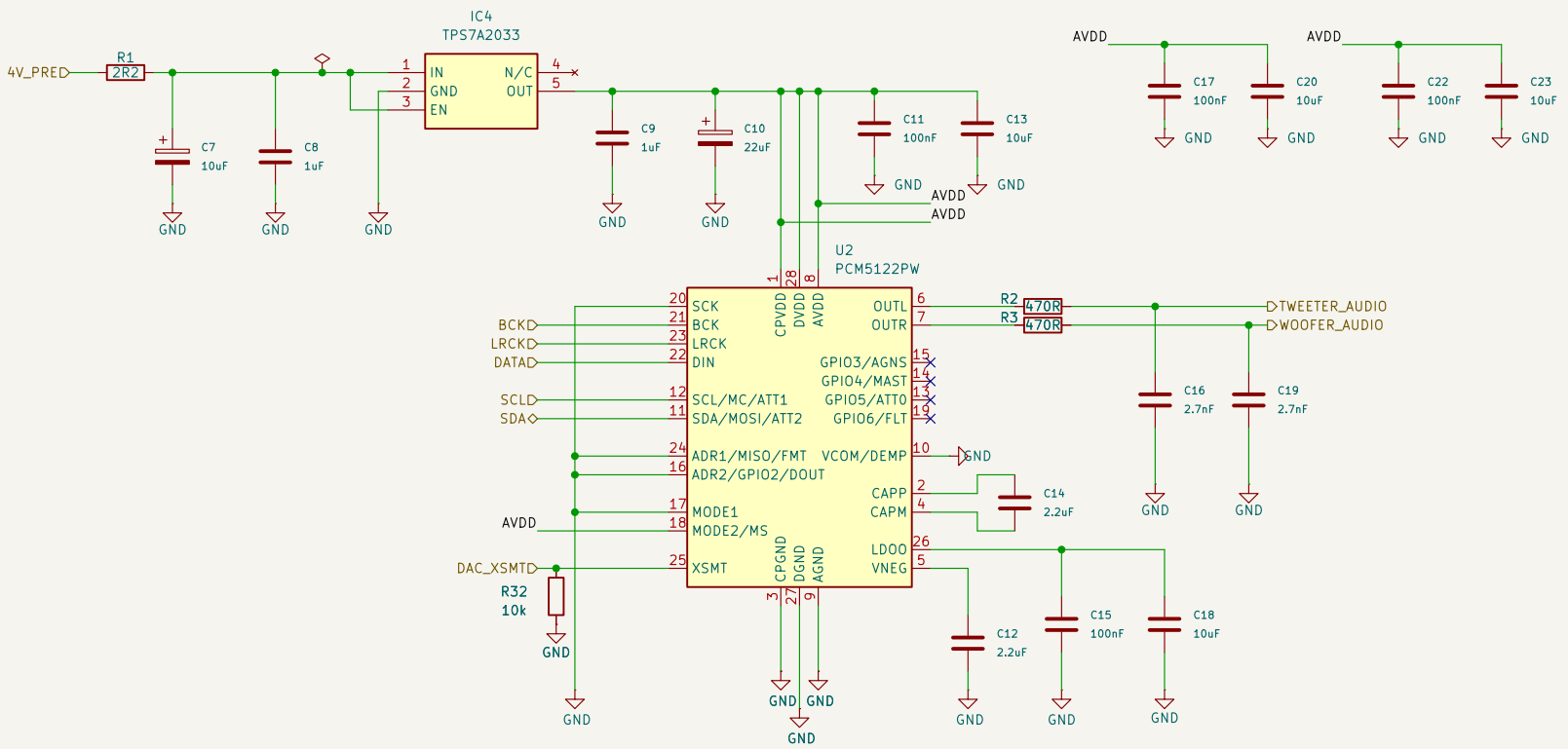
Schematic completion only - PCB unchanged

Sheet: /ESP32/
File: ESP32.kicad_sch

Title: ESP32-WROVER-IE - retained firmware GPIOs

Size: A4	Date: 2026-10-23	**Rev: Pre-order B**
KiCad E.D.A. 10.0.5		Id: 2/5

PCM5122 – I2C VOLUME / MCLK-LESS I2S



MODE1=0, MODE2=1; ADR2=ADR1=0 -> I2C 0x4C (7 bit). SCK grounded.
BCK -> PLL reference; use 64fs BCK and I2S 32-bit slots. Enable normal clock-error handling.
P0 R13 SRCREF=BCK (0x10). P0 R61/R62 = channel digital volume; ramp together for master volume.
OUTL -> tweeter / J17 (TPA left bridge); OUTR -> woofer / J16 (TPA right bridge).
R2/R3 + C16/C19 retain 470R/2n7 C0G output filtering; amp input pads limit full-scale level.
XSMT pulled low by R32 until GPIO18 releases mute. Keep volume muted throughout setup.
VCOM grounded in default VREF mode. LDOO decoupled; unused GPIO3-6 are no-connects.
Clock loss automatically powers down DAC/line drivers. Valid but incorrect DSP data still needs limiting.
All grounds share GND; route analogue return to source, away from amplifier switching currents.

Schematic completion only – PCB unchanged

Sheet: /DAC/
File: DAC.kicad_sch

Title: PCM5122 – L tweeter / R woofer

Size: A4	Date: 2026-10-23	**Rev: Pre-order B**
KiCad E.D.A. 10.0.5		Id: 3/5

TPA3126D2 — STEREO BTL / 4 OHM LOADS

20 dB MASTER; AM[2:0]=000 -> 400 kHz nominal. MODSEL=0 -> hybrid mode.
Each CSAD package: winding 1–2, winding 4–3. No assumed flux cancellation.
TI Fig.31 filter: 10u per leg + 680n to GND per leg (four film capacitors total).
ATS-TI1OP-521-C1-R1 over U1; fit checked by user. Top PowerPAD, no PCB EP copper.
Top pad may be grounded (preferred) or floating per TI; never connect it to PVDD.

PVDD
C57 1nF
C56 1nF
C35 220uF
C33 100nF
C36 220uF
C34 100nF
GND

U1
TPA3126D2DADR

PVCC 18
PVCC 19
PVCC 31
PVCC 32
AVCC 17
GVDD 7
RINP 4
RINN 5
LINP 10
LINN 11
MODSEL 1
SDZ 2
PLIMIT 6
GAIN_SLV 8
MUTE 12
AM2 13
AM1 14
AM0 15
SYNC 16

OUTPR 29
OUTNR 27
BSNR 26
BSPR 30
BSPL 24
BSNL 20
OUTPL 23
OUTNL 21
FAULTZ 3
GND 9
GND 22
GND 25
GND 28

C28 1uF
AMP_RINP
AMP_RINN
AMP_LINP
AMP_LINN
AMP_SDZ
PLIMIT
GAIN_SLV
AMP_MUTE
C25 220nF
C24 220nF
C26 220nF
C27 220nF
SW_WP
SW_WN
SW_TP
SW_TN
AMP_FAULT_N

L2 10uH
L3 10uH
C37 680nF
C38 680nF
C39 680nF
C40 680nF
TW_FILTER_P
TW_PROTECTED_P

J16 WOOFER / CH1
J17 TWEETER / CH2
C32 10nF
R6 3R3
C54 10nF
R20 3R3
C59 10nF
R34 3R3
C60 10nF
R40 3R3

JP1 Jumper

C21 47uF

SDZ default LOW; MUTE default HIGH. 10k/100n command filtering
limits slew below TI 10 V/ms. SDZ asserted low is the primary safe state.
MUTE command HIGH=mute, LOW=play. Wait >=50 ms after SDZ high
before unmuting. FAULTZ open-drain -> GPIO35, 100k pull-up.
No automatic retry link: log fault, mute, cycle SDZ only after diagnosis.

SAFE STARTUP / FAULT

3V3-ESP
AMP_ENABLED
R21 10K
R25 100K
C30 100nF
R30 100K
R29 100K
R26 10K
C31 100nF
AMP_MUTE_CMD
AMP_MUTE
AMP_FAULT_N

15 W / CHANNEL TARGET AT 16 V

PLIMIT: 3V3-ESP -> R19 10k / C29 1u RC; no lower divider resistor.
TI peak-rail model ~19W unclipped / 4 ohm nominal.
Headroom for 15W; not an exact 15W protection limit.
Input pad 1k2 / 1k5 plus DAC 470R -> 0.466 gain.
15W / 4 ohm = 7.75Vrms: calibrate DSP per channel.
Power derates below 16V; measure clipping / thermals.

R19 10K
C29 1uF
R18 5K6
R13 1K5
R12 1K2
C1 1uF
C2 1uF
R16 820R
WOOFER_AUDIO

ONBOARD TWEETER CAPACITOR / JP1 STEREO BYPASS

C21 = Nichicon UES1H470MPM, 47u / 50V bipolar, 10 x 20 mm.
Onboard in series with filtered tweeter positive -> J17 pin 1.
JP1 OPEN: capacitor active; JP1 LINKED: bypass for stereo use.
Use a short insulated wire link rated >=3 A; change only with power off.
Neither speaker terminal is ground. Without cap or link, output is open.
Nominal 4 ohm pole ~847 Hz; actual response follows tweeter impedance.
Account for capacitor phase/response in the DSP crossover.
Heatsink and C21 fit checked by user (23 Oct 2026); bench thermal test remains.

OUTPUT RC: C32/C54/C59/C60 = 10n C0G / 50V, FITTED.
R6/R20/R34/R40 = 3R3 / 0.5W pulse-proof, FITTED.
Post-filter branches to GND; check resistor heating on bench.
Route source-side input reference returns to the DAC GND.
Power loops must not share those return traces.

R15 1K5
R14 1K2
C6 1uF
C8 1uF
R17 820R
TWEETER_AUDIO

WOOFER / J16: DAC RIGHT -> AMPLIFIER RIGHT BRIDGE

TWEETER / J17: DAC LEFT -> AMPLIFIER LEFT BRIDGE

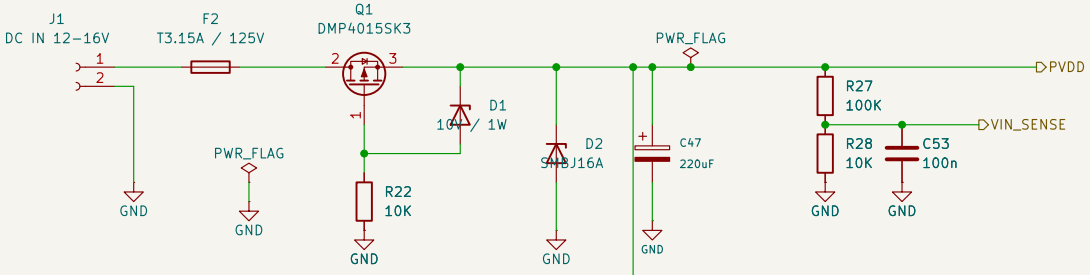
Schematic completion only — PCB unchanged

Sheet: /AMP/
File: Amp.kicad_sch

Title: TPA3126D2 — stereo BTL / C21 bypass

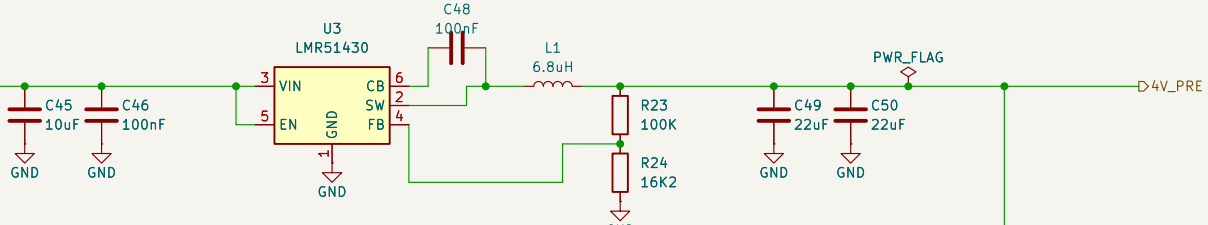
Size: A3	Date: 2026-10-23	Rev: Pre-order B
KiCad E.D.A. 10.0.5		Id: 4/5

INPUT: 12–16 V DC ONLY



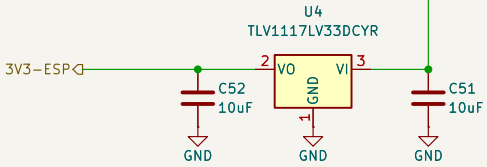
J1 pin 1 = +DC; pin 2 = GND. F2: T3.15A. Q1 reverse polarity; D1 clamps |VGS|.
D2 SMBJ16A: 16 V standoff, 26 V rated pulse clamp. Not a sustained overvoltage regulator.
C47 input bulk; C35/C36 are local amplifier bulk. Use a regulated, current-limited SELV supply.
R27/R28: PVDD/11 -> GPIO36; 100n filtering. ADC1 works while Wi-Fi is active.

4.30 V PRE-REGULATOR



U3 retained: 3 A, 500 kHz forced PWM. 0.6 x (1 + 100k/16k2) = 4.304 V.
L1 6u8: ripple ~0.93 A p-p at 16 V, nominal 500 kHz. Local ceramics at VIN/SW.
C45 10u / 50V X5R: ~3u typical at 16V; estimated VIN ripple ~0.13Vpp at 1A output.
Feeds both 3.3 V LDOs. Do not substitute the PFM LMR51430X version.
VIN_SENSE is advisory: mute/shut down early during a falling input; verify hold-up on bench.

3.3 V ESP32 LDO



TLV1117LV33: 1 A; tab = output. Retained topology, reduced input voltage.
At 0.25 A: ~0.25 W; at 0.5 A: ~0.50 W; at 1 A: ~1.00 W.
Use copper heat spreading. C52 10u ceramic plus C55 22u at ESP32.
TPS7A2033 on DAC sheet supplies only the DAC.

Schematic completion only – PCB unchanged

Sheet: /POWER/
File: Power.kicad_sch

Title: Power – protected input, 4.30 V, 3.3 V

Size: A4	Date: 2026-10-23	**Rev: Pre-order B**
KiCad E.D.A. 10.0.5		Id: 5/5